

CSE4107
Class Test 2, Spring 2019

Marks: 10

Time: 25 min.

1. Write a short note on 'Successor function'. (2)
2. What are the main differences between A* search and Greedy Best-First search? (3)
3. Explain the positive side of Hill-climbing local search. (2)
4. Elaborate the concept of New Population in connection with Genetic algorithms. (3)

1. Write a short note on 'State Space'. (2)
2. Illustrate the concept of global maximum in connection with Hill-climbing local search. (3)
3. What do you know about Priority Queues? (2)
4. Explain the importance of Crossover used in Genetic algorithms. (3)

1. Explain the positive side of Greedy Best-First search. (3)
2. Justify the naming of the strategy called Hill-climbing Local Search. (3)
3. How the parent generation is formed in Genetic algorithms? (2)
4. Explain the positive side of Greedy Best-First search. (2)

1. Illustrate the concept of consistency of heuristic functions with a concise example. (3)
2. Why a genetic algorithm is called a stochastic beam search? (2)
3. Explain the positive and negative sides of A* search. (3)
4. Write a short note on First-choice Hill-climbing search. (2)

1. Write a short note on 'Local search'. (2)
2. Explain the major steps of A* search. (3)
3. Write a brief essay on heuristic functions. (3)
4. How evolution is modeled in Genetic algorithms? (2)

1. Distinguish between blind and informed search. (2)
2. Describe the major steps of a typical Hill-climbing local search algorithm. (3)
3. How reproduction is modeled in Genetic algorithms? (2)
4. Illustrate the process of updating the Possible Path in Greedy Best-first search. (3)

1. How do you explain a Goal test in search problems? (2)
2. Explain three important decisions about A* search. (3)
3. Write a brief essay on 'Some real problems require Hill-climbing local search'. (3)
4. Elaborate the concept of Mutation in connection with Genetic algorithms. (2)

1. Write a short note on 'Optimal solution of a search problem'. (2)
2. Illustrate the odds of the environment of Hill-climbing local search problems. (3)
3. What do you know about the evaluation function of Greedy best-first search? (2)
4. Narrate the ideas that stand behind Genetic algorithms. (3)

1. Describe the conditions under which heuristic functions are considered good. (3)
2. Explain three of the most important features of Hill-climbing Local Search. (3)
3. How the new generation is formed in Genetic algorithms? (2)
4. Mark the negative points in Greedy Best-First search. (2)

1. Explain the necessity of heuristic functions for problem solving. (2)
2. Narrate those major steps of a typical genetic algorithm that are repeatedly executed. (3)
3. What are the problems with A* search, and how those can be addressed? (3)
4. Write a short note on Random restart Hill-climbing search. (2)

1. Write a short note on 'Local maximum'. (2)
2. Narrate the major steps of A* search that are repeatedly executed. (3)
3. Write a brief essay on Consistency of heuristic functions. (3)
4. How Crossover is executed in Genetic algorithms? (2)

1. Write a short note on 'Admissibility of heuristic functions'. (2)
2. Describe the repeatedly executed major steps of a typical Hill-climbing local search. (3)
3. How randomness is applied in Genetic algorithms? (2)
4. Explain the positive and negative sides of Greedy Best-first search. (3)

1. Write a short note on 'Solution to AI problems'. (2)
2. Why and how a tree of nodes is maintained in Greedy Best-first search? (3)
3. For what type of problems Hill-climbing local search is appropriate? (2)
4. Elaborate the concept of Population in connection with Genetic algorithms. (3)

1. Write a short note on 'Path cost of AI problems'. (2)
2. Illustrate the concept of local maxima in connection with Hill-climbing local search. (3)
3. What do you know about the evaluation function of A* search? (2)
4. Narrate the importance of Crossover and Mutation used in Genetic algorithms. (3)

1. Illustrate the conditions under which heuristic functions are considered good. (3)
2. Justify the naming of the strategy called Hill-climbing Local Search. (3)
3. How the new generation is formed in Genetic algorithms? (2)
4. Explain the negative side of A* search. (2)

1. Illustrate the concept of consistency of heuristic functions with a concise example. (3)
2. Why a genetic algorithm is called a local search? (2)
3. Explain the positive and negative sides of Greedy Best-first search. (3)
4. Write a short note on 'Stochastic Hill-climbing search'. (2)

1. Write a short note on 'Hill-climbing search'. (2)
2. Narrate the major steps of A* search that are repeatedly executed. (3)
3. Illustrate the concept of heuristic functions. (3)
4. How Crossover is executed in Genetic algorithms? (2)

1. Distinguish between blind and informed search. (2)
2. Describe the repeatedly executed major steps of a typical Genetic algorithm. (3)
3. Explain the positive side of Hill-climbing local search. (2)
4. Illustrate the process of updating the Possible Path in A* search. (3)